

Chapter 4: Regression

School of Mathematics and Computational Sciences
University of Prince Edward Island

Instructor: Ali Raisolsadat

Part I: Exploring Data

Why Regression Matters

- Chapter 3 introduced scatterplots and correlation as tools for describing the direction and strength of a linear association between two quantitative variables.
- Regression goes one step further. It gives an equation for a fitted line that can be used to describe how a response variable tends to change as an explanatory variable changes.
- Regression is important because it connects visual patterns, formulas, prediction and interpretation .

From Correlation to Regression

Correlation and regression are related but they answer different questions.

Idea	Correlation	Regression
Main purpose	Describes the direction and strength of a linear association	Describes the fitted relationship and predicts the response
Roles of variables	The two variables are treated symmetrically	One variable is explanatory and one is the response
Units	Unitless	The slope has response units per explanatory-variable unit
Typical question	How strong is the linear association	What response is predicted for a given value of x

The Roles of the Variables

- A regression analysis begins by assigning roles to the variables.
 - The explanatory variable x is used to explain or predict.
 - The response variable y is the outcome being described or predicted.
- These roles come from the scientific question rather than from the order of the columns in a dataset.
- A fitted line describes an average pattern. Two individuals with the same value of x can still have different observed responses.

Regression as a Model

- A fitted regression line summarizes a linear pattern in the observed data.
- The fitted line does not guarantee that every point lies on the line.
- It does not establish that changes in the explanatory variable cause changes in the response variable.

The Anatomy of a Straight Line

- A straight line relating x to y has the form

$$y = a + bx$$

- The slope b describes the change in y when x increases by one unit.
- The intercept a is the value of y when $x = 0$.

Example

- Consider the straight line

$$y = 3 + 2x$$

- For this line

$$a = 3 \quad \text{and} \quad b = 2$$

- The line crosses the y -axis at $(0, 3)$.
- Each one-unit increase in x produces a two-unit increase in y .

x	0	1	2	3
$y = 3 + 2x$	3	5	7	9

How the Slope and Intercept Are Read

For the straight line

$$y = a + bx$$

the intercept and slope describe different features of the line.

- The **intercept** is a . Setting $x = 0$ gives

$$y = a + b(0) = a$$

Therefore, the line crosses the y -axis at the point $(0, a)$.

- The **slope** is b . It can also be written as

$$b = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x}$$

- If $b > 0$, the line rises from left to right. If $b < 0$, the line falls from left to right. If $b = 0$, the line is horizontal.

From an Algebraic Line to a Regression Line

A regression line uses the same algebraic form but the predicted response is written as $\hat{y}$.

$$\hat{y} = a + bx$$

The notation matters.

- y is an observed response taken from the data
- $\hat{y}$ is the response predicted by the fitted line

This distinction becomes very important when residuals are introduced.

Reading a Regression Equation

Question

A greenhouse study uses the fitted equation

$$\widehat{\text{height}} = 4.5 + 1.8(\text{hours of light})$$

to predict final plant height in centimetres from daily light exposure in hours.

What does the slope mean? What height does the model predict for a plant receiving 6 hours of light per day?

Additional Example: Solution

- The slope is 1.8 centimetres per hour of light.
- According to the fitted linear model, a one-hour increase in daily light exposure is associated with an increase of 1.8 centimetres in predicted final height.
- The intercept is 4.5 centimetres. It is the predicted height when the explanatory variable equals zero. Whether that interpretation is useful depends on the scientific setting.
- For 6 hours of light, the predicted height is

$$\widehat{\text{height}} = 4.5 + 1.8(6)$$

$$\widehat{\text{height}} = 15.3 \text{ cm}$$

- The fitted model predicts a final height of 15.3 cm for a plant receiving 6 hours of light per day.

Example 4.1: Predicting Fat Gain

Question

- A study followed 16 young adults who overate for eight weeks.
- The explanatory variable was the change in nonexercise activity (NEA) measured in Calories. The response variable was fat gain measured in kilograms.
- How can the fitted regression line be used to predict fat gain for a person whose NEA increases by 400 Calories?

Volunteer	NEA Change (Calories)	Fat Gain (kg)
1	-94	4.2
2	-57	3.0
3	-29	3.7
4	135	2.7
5	143	3.2
6	151	3.6
7	245	2.4
8	355	1.3

Volunteer	NEA Change (Calories)	Fat Gain (kg)
9	392	3.8
10	473	1.7
11	486	1.6
12	535	2.2
13	571	1.0
14	580	0.4
15	620	2.3
16	690	1.1

Example 4.1: Explanation

- The fitted least-squares regression line for these data is

$$\widehat{\text{fat gain}} = 3.505 - 0.00344(\text{NEA change})$$

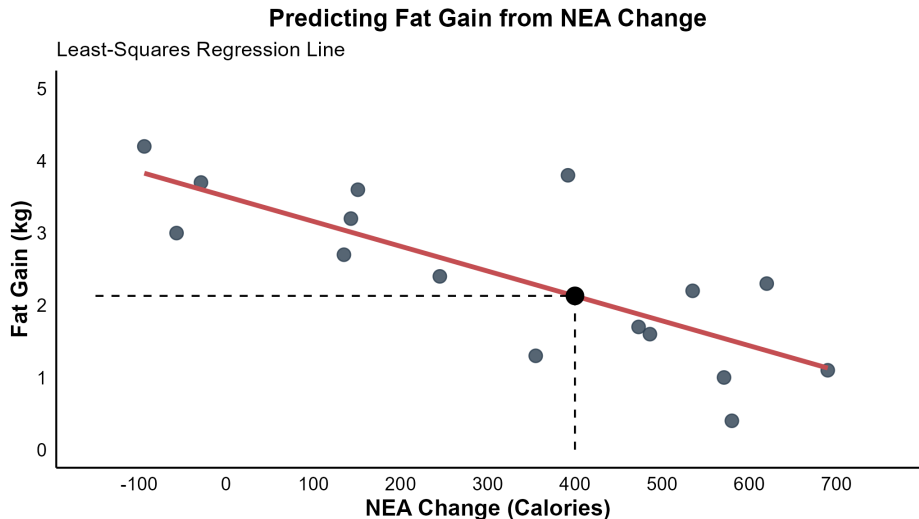
- A prediction at an NEA change of 400 Calories is obtained by substituting $x = 400$ into the fitted equation.
- The value 400 lies inside the observed range of NEA changes, so this prediction is an interpolation rather than an extrapolation.
- The predicted fat gain is

$$\widehat{\text{fat gain}} = 3.505 - 0.00344(400)$$

$$\widehat{\text{fat gain}} \approx 2.13 \text{ kg}$$

- The value 2.13 kg is the response predicted by the fitted line for a person whose NEA increases by 400 Calories.

Example 4.1: Scatterplot and Fitted Line



Reading the Context

- The slope is -0.00344 kg per Calorie.
- Within the range of NEA values represented by the data, the fitted model predicts that fat gain decreases by about 0.00344 kg for each additional Calorie of NEA change.
- A 100-Calorie increase in NEA corresponds to a predicted decrease of about 0.344 kg in fat gain according to the fitted line.

$$\widehat{\text{fat gain}} = 3.505 - 0.00344(100) = 3.505 - 0.344 = 3.161$$

The Meaning of the Intercept

- The intercept is 3.505 kg. It represents the predicted fat gain when NEA change is zero.

$$\widehat{\text{fat gain}} = 3.505 \text{ kg when NEA change} = 0$$

- In this example, the value $x = 0$ lies inside the observed range, so the intercept has a meaningful interpretation.
- In other applications, the intercept may be mathematically necessary while having little practical meaning.

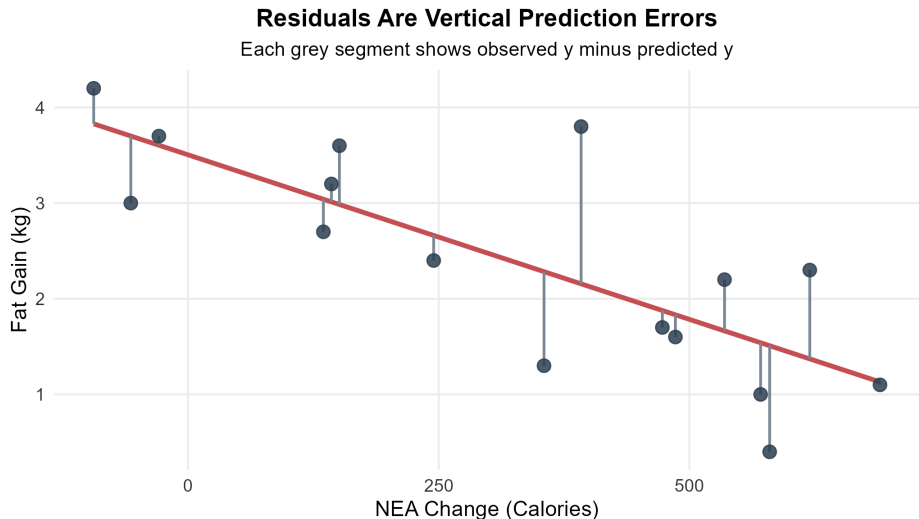
Least-Squares Regression

- A line drawn by eye is subjective. Different people could draw slightly different lines through the same scatterplot.
- Least-squares regression provides a precise mathematical rule for choosing one line.
- For an observation (x_i, y_i) , the fitted line gives a predicted response $\hat{y}_i$. The residual is the signed vertical difference between the observed response and the predicted response.

$$e_i = y_i - \hat{y}_i$$

- A residual compares what was observed with what the fitted line predicted.
 - A positive residual means the observed point lies above the fitted line
 - A negative residual means the observed point lies below the fitted line
 - A residual close to zero means the fitted line predicted that observation well

Residual Plot Illustration



Residuals from the NEA Study

Question

Volunteer 8 had an NEA change of 355 Calories and an observed fat gain of 1.3 kg. What fat gain does the fitted line predict for this volunteer and what is the residual?

Recall that

$$\widehat{\text{fat gain}} = 3.505 - 0.00344(\text{NEA change})$$

The fitted value is

$$\hat{y} = 3.505 - 0.00344(355)$$

$$\hat{y} \approx 2.28 \text{ kg}$$

The residual is

$$e = y - \hat{y} = 1.3 - 2.28$$

$$e \approx -0.98 \text{ kg}$$

The residual is negative. So this point lies below the fitted line.

Why Are the Residuals Squared

The least-squares line is the line that minimizes the sum of squared residuals.

$$\text{Sum of squared residuals} = \sum e_i^2$$

Squaring serves two purposes.

- Positive and negative residuals do not cancel out
- Large prediction errors are penalized more strongly than small ones

Least-Squares Regression Formulas

$$b = r \frac{s_y}{s_x} \quad (\text{slope})$$

$$a = \bar{y} - b\bar{x} \quad (\text{intercept})$$

$$\hat{y} = a + bx \quad (\text{line equation})$$

$$e = y - \hat{y} \quad (\text{residuals})$$

These formulas connect the regression coefficients to the summary statistics from Chapter 3.

What the Slope Formula Tells Us

The slope formula

$$b = r \frac{s_y}{s_x}$$

shows that the slope depends on three ingredients.

- The sign and strength of the linear association through r
- The spread of the response variable through s_y
- The spread of the explanatory variable through s_x

A positive correlation gives a positive slope. A negative correlation gives a negative slope.

Example 4.2: Calculating the Line from Summary Statistics

Question

For the NEA and fat-gain data, the summary statistics are

$$\bar{x} = 324.75$$

$$\bar{y} = 2.3875$$

$$s_x = 261.46$$

$$s_y = 1.1207$$

$$r = -0.8021$$

Use these summary statistics to calculate the least-squares regression line.

Example 4.2: Calculate the Slope and Intercept

Using the slope formula

$$b = r \frac{s_y}{s_x}$$

$$b = (-0.8021) \left(\frac{1.1207}{261.46} \right)$$

$$b \approx -0.00344$$

The fitted line has a negative slope which agrees with the negative association seen in the scatterplot.
Using the intercept formula

$$a = \bar{y} - b\bar{x}$$

$$a = 2.3875 - (-0.00344)(324.75)$$

$$a \approx 3.505$$

The least-squares regression line is therefore

$$\widehat{\text{fat gain}} = 3.505 - 0.00344(\text{NEA change})$$

Example 4.2: Residual Calculations

A residual is the difference between the observed response and the fitted response:

$$e_i = y_i - \hat{y}_i$$

i	x_i	y_i	$\hat{y}_i$	e_i
1	-94	4.2	3.829	0.371
2	-57	3.0	3.701	-0.701
3	-29	3.7	3.605	0.095
4	135	2.7	3.041	-0.341
5	143	3.2	3.013	0.187
6	151	3.6	2.985	0.615
7	245	2.4	2.662	-0.262
8	355	1.3	2.283	-0.983

i	x_i	y_i	$\hat{y}_i$	e_i
9	392	3.8	2.156	1.644
10	473	1.7	1.877	-0.177
11	486	1.6	1.833	-0.233
12	535	2.2	1.664	0.536
13	571	1.0	1.540	-0.540
14	580	0.4	1.509	-1.109
15	620	2.3	1.371	0.929
16	690	1.1	1.130	-0.030

A positive residual means the observed value lies above the regression line. A negative residual means the observed value lies below the line. Using the unrounded fitted values

$$\sum_{i=1}^{16} e_i = 0.001$$

Calculating the Least-Squares Line in R

```
# Enter the observed data
nea_change <- c( -94, -57, -29, 135, 143, 151, 245, 355, 392, 473, 486, 535, 571, 580, 620, 690)
fat_gain <- c(4.2, 3.0, 3.7, 2.7, 3.2, 3.6, 2.4, 1.3, 3.8, 1.7, 1.6, 2.2, 1.0, 0.4, 2.3, 1.1)

# Calculate the required summary statistics
mean_x <- mean(nea_change)
mean_y <- mean(fat_gain)

sd_x <- sd(nea_change)
sd_y <- sd(fat_gain)

correlation <- cor(nea_change, fat_gain)

# Calculate the slope and intercept
slope <- correlation * (sd_y / sd_x)
intercept <- mean_y - slope * mean_x

> intercept      slope
>  3.505123 -0.003441
```

Therefore, the least-squares regression line is $\hat{y} = 3.5051 - 0.003441x$

R Code: Fitting the Least-Squares Line with `lm()`

R provides the `lm()` function for fitting a least-squares regression line.

Fit the regression model

```
# Fit fat gain as a function of NEA change
model <- lm(fat_gain ~ nea_change)

# Display the estimated coefficients
round(coef(model), digits = 6)

> (Intercept)  nea_change
>    3.505123   -0.003441
```

Reading an R model formula

`response ~ explanatory variable`

- The variable on the left of `~` is the response variable: `fat_gain`.
- The variable on the right is the explanatory variable: `nea_change`.
- The first coefficient is the intercept, and the second coefficient is the slope.

Therefore, R produces the regression line

$$\widehat{\text{fat gain}} = 3.5051 - 0.003441(\text{NEA change})$$

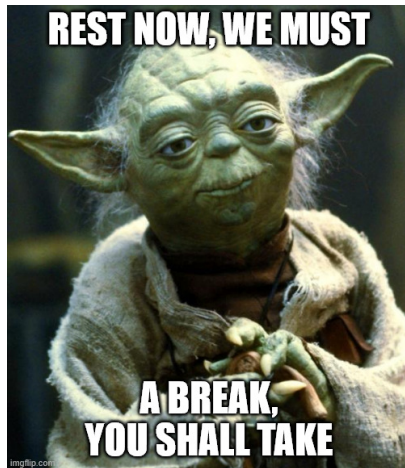
R Code: Fitted Values and Residuals

```
# Fitted responses and residuals from the linear model
fitted_values <- fitted(model)
residual_values <- residuals(model)

# A standard summary of the fitted linear model
summary(model)
```

At this stage, the fitted coefficients and R^2 are the most relevant parts of the output.

Break Time



Facts About Least-Squares Regression

Several mathematical properties help explain how a fitted line behaves.

- The roles of x and y are not interchangeable
- The sign of the slope always matches the sign of the correlation
- The least-squares line always passes through $(\bar{x}, \bar{y})$
- The coefficient of determination R^2 summarizes how well the line accounts for variation in the response

Slope and Correlation

The slope and correlation are connected by the formula

$$b = r \frac{s_y}{s_x}$$

The sign of the slope always matches the sign of r .

- If $r > 0$, then $b > 0$
- If $r < 0$, then $b < 0$
- If $r = 0$, then $b = 0$

The Roles of x and y

The least-squares line for predicting y from x is generally different from the least-squares line for predicting x from y .

This difference occurs because the regression line minimizes vertical prediction errors in the response variable.

Regression is therefore directional in a way that correlation is not.

The Point of Averages

The least-squares regression line always passes through the point of averages.

$$(\bar{x}, \bar{y})$$

This fact is built into the intercept formula

$$a = \bar{y} - b\bar{x}$$

It provides a useful algebraic check when the fitted line is calculated by hand.

A Second View of R^2

The variation in the observed response values can be divided into two parts: variation explained by the regression line and variation left in the residuals.

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

Total variation in the response,

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

Variation explained by the regression line,

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2$$

Variation left unexplained

A Second View of R^2

For a least-squares regression line with an intercept

$$SST = SSR + SSE$$

Therefore, the coefficient of determination can be written in two equivalent ways:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

- $\frac{SSR}{SST}$ is the proportion of total variation explained by the regression line.
- $\frac{SSE}{SST}$ is the proportion of total variation left unexplained.
- For example, $R^2 = 0.64$ means that the linear model explains 64% of the variation in the response variable. The remaining 36% is left in the residuals.

Example 4.3: Interpreting R^2

Question

Suppose a fitted linear model has

$$R^2 = 0.75$$

How should this value be interpreted?

- An R^2 value of 0.75 means that 75% of the observed variation in the response variable is accounted for by the fitted linear relationship with the explanatory variable.
- The remaining 25% is not accounted for by the fitted line and remains in the residual variation.
- This interpretation describes the quality of fit. It does not imply causation.

Outliers and Influential Observations

A single unusual observation can affect correlation and least-squares regression in important ways.

Three related ideas should be distinguished.

- An outlier has an unusually large residual
- A high-leverage point has an unusual value of x
- An influential point changes the fitted line noticeably when it is included or removed

Examples 4.4 and 4.5: The Empathy Study

Question

An empathy study measures brain activity and empathy score for 16 women. Subject 16 has an unusually large empathy score.

How does this point affect the correlation and the least-squares regression line?

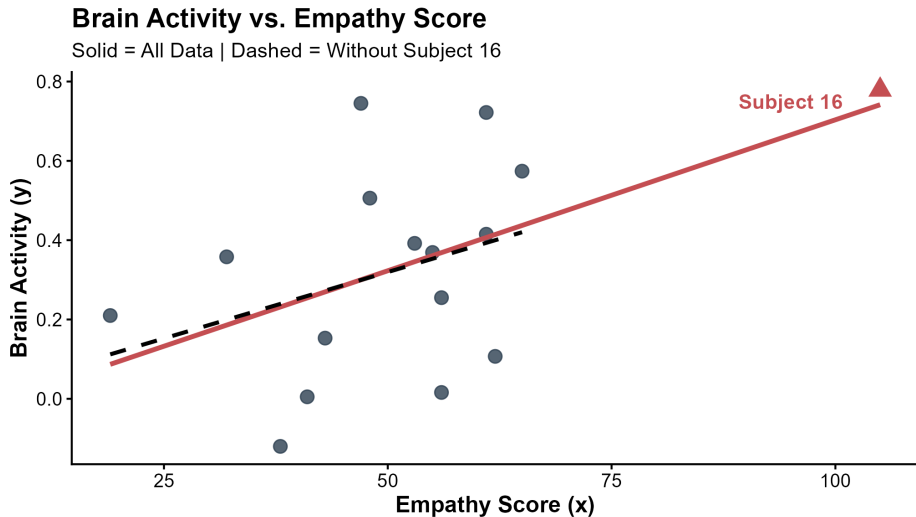
- Subject 16 changes the correlation substantially.

$$r = 0.515 \text{ with Subject 16}$$

$$r = 0.331 \text{ without Subject 16}$$

- The fitted regression line also changes, although less dramatically than the correlation.
- This example shows that a point can be influential for both correlation and regression, with different levels of impact.

Empathy Study Plot



A Hypothetical High-Leverage Point

A point with an extreme x value has leverage because it can pull the fitted line toward itself. If a high-leverage point also has a large residual, it can become highly influential. This is why unusual observations should be examined carefully before a fitted line is interpreted.

Cautions About Correlation and Regression

A fitted line should always be interpreted in light of the data and the scientific question.

- Correlation and simple linear regression describe linear patterns
- Unusual observations can change the correlation and the fitted line
- Predictions outside the observed range require caution
- A lurking variable can help explain an observed association
- Association by itself does not imply causation

Linear Relationships

A scatterplot should be examined before a simple linear model is interpreted.

Curvature, separate clusters or changing spread can all indicate that a straight line is an inadequate summary.

R or other statistical software can always produce a correlation coefficient and a fitted line, but those outputs are only useful when the relationship is reasonably linear.

Interpolation and Extrapolation

A prediction within the observed range of x values is called interpolation.

A prediction outside the observed range is called extrapolation.

Algebra allows the fitted line to be extended indefinitely, but the data do not show whether the same relationship continues beyond the observed range.

Additional Example: Interpolation versus Extrapolation

Question

A fertilizer study observes daily doses from 0 to 5 millilitres and fits a linear regression model over that range.

Which prediction is better supported by the observed data: a prediction at 3 millilitres or a prediction at 100 millilitres?

- The prediction at 3 millilitres is better supported because it is an interpolation.
- The prediction at 100 millilitres is an extrapolation far beyond the observed data.
- The fitted line contains no information about whether the biological relationship remains linear from 5 to 100 millilitres.

Lurking Variables

A lurking variable is a variable that is not included as an explanatory or response variable in the analysis but may help explain the observed association.

For example, an observational study might find that coffee consumption is positively associated with risk of heart attack. In this case, smoking could be a lurking variable if it is related to both coffee consumption and heart attack risk.

Association Does Not Imply Causation

- Regression and correlation describe associations. They do not by themselves establish cause and effect.
- A causal conclusion requires more than a strong correlation, a steep slope or a large R^2 . The study design matters.
- A well designed randomized experiment can provide stronger evidence for a causal effect because random assignment helps balance lurking variables across treatment groups.
- Observational studies generally require more caution because unmeasured confounding can remain even after statistical adjustment.

A Consistent Regression Workflow

A useful first-year workflow is to interpret regression in the same order every time.

1. Identify the explanatory variable and the response variable
2. Examine the scatterplot for linearity and unusual observations
3. Write and interpret the fitted line
4. Calculate or interpret a prediction when appropriate
5. Use residuals and R^2 to assess the fit
6. Consider limitations such as extrapolation, lurking variables and causation